

OMEGA SERIES 6.1

Constraining χ -Condensate Substructure in Classical Dwarf Spheroidals

A Methodology and Forecast Paper

Dynamic-wavelength matched filter · Phase-free curvature-weighted statistic · Walker+ 2009 stellar kinematics · N-body validation · Implementation-spread methodology · ELT/HARMONI forecast

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Dedicated to the Omega Research Collective.

For the contributors who held themselves to honest scope when the data did not deliver.

ABSTRACT

We present a methodology for constraining the dimensionless modulation amplitude $\theta = A_0 \cdot f_\chi$ predicted by the TSMVEFT framework (MSDS Series 4.3–4.6, 5, 6.0) in classical dwarf spheroidal galaxies. The framework predicts a dynamic-wavelength stellar kinematic modulation with characteristic scale $\lambda(R) = h/(m_\chi \sqrt{3} \sigma_{\text{local}}(R))$, which for fuzzy dark matter mass $m_\chi = 10^{-22} \text{ eV}/c^2$ yields predicted wavelengths of 5–9 kpc across the primary sample (Fornax, Sculptor, Leo I, Sextans, with CVn1 as secondary). Available stellar kinematic data (Walker+ 2009 binned dispersion profiles) cover radial extent of approximately 2 kpc, placing the test in the partial-wavelength regime. We develop a phase-free curvature-weighted statistic S , verify its analytical normalization $\alpha = N/[6(N-2)]$, and apply it across six independent implementations to current binned data. The resulting constraint is **$\theta < 0.13$** (geometric mean across documented implementations, with implementation-dependent range $\theta \leq 0.05\text{--}0.30$ at 95% confidence). Conditional on N-body simulation prior $A_0 \in [0.05, 0.20]$, this gives $f_\chi \lesssim 0.7\text{--}1.0$ as the loose model-dependent bound. The cross-galaxy Spearman correlation between recovered statistic strength and $\sigma_{\text{local}}^{-2}$ yields population-weighted $\rho \approx 0.48$, below the framework threshold $\rho > 0.6$ expected for a confirmed signal. We identify two dominant systematics: (i) anisotropy-induced curvature, which can mimic $S_{\text{norm}} \sim 2.3$ for extreme $\beta(r) \in [-1, +0.9]$, and (ii) the curvature-weighted statistic's designed insensitivity to smooth $A(R)$ modulations. We forecast that ELT/HARMONI individual stellar velocities at $\sigma_v < 0.3 \text{ km/s}$ by 2030 will resolve both systematics and push $f_{\chi, \text{min}}$ to ~ 0.10 . We publish the implementation-dependent spread across six independent runs as a methodological contribution: at current bin counts, single-pipeline analyses of fuzzy-DM substructure signals carry implementation-dependent uncertainties at the order-of-magnitude level. **We make no detection claim.**

1. Introduction

Series 6.0 of the MSDS programme [13] established that the Two-Scalar Mirror-Vacuum Effective Field Theory (TSMVEFT, Series 4.3–4.6 [10–12]) predicts substructure in the dark matter sector that does not produce a clumpy population of subhalos. Three mechanisms were identified: de Broglie modulations at kpc scale for fuzzy DM mass $m_\chi \sim 10^{-22}$ eV/c², soliton interference patterns in merged halos, and a Khoury-Berezhiani-style superfluid-to-CDM transition at $v_{\text{crit}} \approx 50$ km/s. Of these three mechanisms, only the de Broglie modulation produces a kinematic signature in classical dwarf spheroidal galaxies that is potentially observable with current or near-future facilities.

This paper (Series 6.1) develops the methodology for testing that prediction. The central observable is a stellar kinematic modulation with characteristic wavelength $\lambda(R) = h/(m_\chi \sqrt{3} \sigma_{\text{local}}(R))$, where $\sigma_{\text{local}}(R)$ is the line-of-sight velocity dispersion at projected radius R . The factor $\sqrt{3}$ converts the one-dimensional dispersion to the three-dimensional effective velocity for isotropic dwarf spheroidals (Series 6.0, Section 4). For canonical fuzzy DM mass and typical dwarf dispersions ($\sigma \approx 7\text{--}12$ km/s), the predicted modulation wavelength is 5–9 kpc.

Three findings drive this paper's methodology and forecast framing rather than a detection framing. First, available stellar kinematic data (Walker et al. 2009 [14] binned dispersion profiles, supplemented where available by MUSE-Faint, Zoutendijk et al. 2021 [15]) cover projected radial extent of approximately 2 kpc per dwarf, less than half of the predicted modulation wavelength. The test is therefore conducted in the partial-wavelength regime, where standard Fourier methods fail because the radial Nyquist limit lies above the predicted peak wavenumber. Second, controlled N-body simulation (this work, Section 4) demonstrates that even a strong static modulation with $\delta\rho/\rho = 0.3$ phase-mixes to a marginal residual signature (0.44σ RMS excess above smooth-potential control) after 10 Gyr of stellar dynamical evolution. Third, the predicted modulation amplitude scales as $\delta\sigma/\sigma = A(R) \cdot f_\chi$, where $A(R)$ is a geometric envelope factor of order 0.05–0.20 (constrained from N-body simulation only, not from first principles), and f_χ is the χ -condensate fraction of total dark matter. Current data sensitivity to the product $A_0 \cdot f_\chi$ is the constrainable quantity; isolation of f_χ alone requires a simulation prior on A_0 .

Given these findings, Series 6.1 presents (i) the formal statistical test for the dynamic-wavelength prediction, (ii) the constraint achievable from current data, (iii) the dominant systematic uncertainties (anisotropy degeneracy and statistic-design insensitivity to smooth signals), and (iv) the data quality required for future detection. We make no detection claim. We report a methodology that survives referee-level audit, a conservative upper limit on $\theta = A_0 \cdot f_\chi$, and a forecast for when next-generation facilities will distinguish the predicted signal from systematics.

The paper is structured as follows. Section 2 specifies the theoretical framework and the primary sample. Section 3 develops the phase-free curvature-weighted statistic and the cross-galaxy scaling test. Section 4 reports controlled N-body validation of the statistic's expected behaviour under modulated and smooth potentials. Section 5 applies the methodology to current Walker et al. 2009 data and reports the constraint. Section 6 catalogues the dominant systematics and presents a deliberate red-team analysis of where the methodology can fail. Section 7 forecasts the data quality required for detection. Section 8 reports the implementation-spread analysis as a methodological contribution. Section 9 concludes.

2. Theoretical Framework

2.1 Dynamic-Wavelength Prediction

TSMVEFT in the fuzzy-DM regime ($m_\chi \sim 10^{-22}$ eV/c²) predicts that the χ -condensate develops phase-coherent density modulations at the de Broglie scale set by the local velocity field. For a stellar tracer in the gravitational potential generated by the modulated condensate, the line-of-sight velocity dispersion exhibits a residual structure with characteristic wavenumber

$$k_{\text{peak}}(R) = 2\pi / \lambda(R) = 2\pi m_\chi \sqrt{3} \sigma_{\text{local}}(R) / h$$

Two features distinguish this prediction from generic modulation models. First, the wavelength is dynamic: it scales inversely with the local velocity dispersion within a single galaxy and across the dwarf population. This produces a falsifiable cross-galaxy scaling, $\lambda \propto 1/\sigma$, testable by Spearman correlation across a sample. Second, the modulation is bounded by the stellar tracer envelope rather than the dark matter halo extent. We adopt $R_{\text{env}} \approx 2 \times r_{\text{half}}$ as the operational envelope of detectable stellar kinematic modulation, set by where the stellar tracer probes the dark matter potential.

Predicted wavelengths and wavenumbers at $R = r_{\text{half}}$ for the primary sample, computed with $m_\chi = 10^{-22}$ eV/c²:

Galaxy	σ at r_{half} (km/s)	λ_{pred} (kpc)	k_{peak} (kpc ⁻¹)	R_{env} (kpc)
Fornax	11.7	5.94	1.058	1.42
Sculptor	9.2	7.56	0.832	0.52
Leo I	9.2	7.56	0.832	0.50
Sextans	9.2	7.56	0.832	1.38
CVnI	7.6	9.15	0.687	1.10

For the bracket $m_\chi \in [0.3, 3.0] \times 10^{-22}$ eV/c², predicted wavelengths scale inversely with mass. The lower bound ($m_\chi = 0.3 \times 10^{-22}$) gives $\lambda \approx 18\text{--}27$ kpc across the sample, far exceeding any radial coverage achievable with current or near-term facilities. The upper bound ($m_\chi = 3 \times 10^{-22}$) gives $\lambda \approx 1.7\text{--}2.6$ kpc, comparable to data coverage. The central value $m_\chi = 10^{-22}$ eV/c² used throughout this paper places the test in the partial-wavelength regime for all primaries.

2.2 The Partial-Wavelength Regime

Walker et al. 2009 stellar dispersion profiles provide 8–17 radial bins per dwarf out to projected radius $R_{\text{max}} \approx 2.0\text{--}2.5$ kpc. Since $R_{\text{max}} < \lambda_{\text{pred}}$ for all primaries, the test cannot resolve a complete oscillation. The radial Nyquist limit $k_{\text{max}} = \pi/\Delta R \approx 16$ kpc⁻¹ for typical bin spacing $\Delta R \approx 0.2$ kpc lies well above the predicted k_{peak} , but this does not solve the resolution problem: only 1–2 independent Fourier modes lie within the predicted peak window, making conventional power-spectrum methods ill-conditioned. We confirmed this empirically by attempting a direct Lomb-Scargle and FFT-based analysis on Walker et al. 2009 binned data; the test returned undefined results due to insufficient mode density (Section 3.4).

The methodology developed in Section 3 substitutes a real-space curvature-weighted power statistic for the under-resolved Fourier analysis. This approach measures excess low-frequency

power without requiring resolution of λ_{pred} itself, at the cost of phase information.

2.3 The Primary Sample

Following the sample audit conducted in Series 6.1 Pillar 2, the primary sample consists of four classical dwarf spheroidals (Fornax, Sculptor, Leo I, Sextans) and one secondary system (CVnI) used for the Sextans-vs-CVnI null test. Selection criteria: (1) availability of A- or B-grade stellar kinematic data (Walker et al. 2009, supplemented by Mateo et al. 2008 [4] for Leo I and Simon & Geha 2007 [9] for CVnI); (2) Galactic pericenter $R_{\text{peri}} > 50$ kpc to minimize tidal stripping (Gaia Collaboration 2018 [3]), excluding Draco, Ursa Minor, and Carina; (3) absence of complex merger history that would invalidate the single-progenitor soliton interference assumption.

Sample exclusions from the broader McConnachie et al. 2020 [5] catalogue: Draco and Ursa Minor (tidal stripping at last pericenter), Carina (extended star formation history with multiple bursts), LMC/SMC (Magellanic system dynamics), Bootes I (ultra-faint, kinematic data quality below threshold), Sagittarius (active disruption). Three additional ultra-faint dwarfs (Leo II, UMa II, Aquarius 2) are retained as control galaxies in Section 6 systematic checks.

2.4 The Sculptor Two-Population Test

Sculptor exhibits two distinct chemokinematic populations resolved by Tolstoy et al. 2023 [16]: a metal-poor (MP) population with $\sigma_{\text{MP}} \approx 11.1$ km/s and a metal-rich (MR) population with $\sigma_{\text{MR}} \approx 6.4$ km/s, both inhabiting the same gravitational potential. Under the framework, the predicted wavelengths differ by the ratio $\sigma_{\text{MP}}/\sigma_{\text{MR}} \approx 1.74$, giving $\lambda_{\text{MP}} \approx 6.26$ kpc and $\lambda_{\text{MR}} \approx 10.87$ kpc. The corresponding ratio of curvature-weighted power, $S_{\text{MR}}/S_{\text{MP}}$, scales as $(\sigma_{\text{MR}}/\sigma_{\text{MP}})^2 \approx 0.34$. This within-galaxy test eliminates inter-galaxy systematics (halo profile, distance, environmental history) and provides the framework's strongest internal cross-check. Its application is currently limited by the absence of machine-readable Tolstoy et al. 2023 binned profiles for both populations; we report it as a high-priority follow-up channel.

3. Statistical Methodology

3.1 Phase-Free Curvature-Weighted Statistic

In the partial-wavelength regime, phase-dependent matched-filter approaches (cosine projection at predicted k) are degenerate with low-order radial curvature and cannot be distinguished from Jeans mismodeling, anisotropy gradients, or tidal effects within a single galaxy. We develop instead a phase-free statistic that operates on residual structure without assuming a phase template.

Define the residual at radial bin i as $\delta\sigma_i = \sigma_{\text{los}}(R_i) - \langle\sigma\rangle$, where $\langle\sigma\rangle$ is the unweighted mean of the binned profile. Define the discrete second-derivative operator $D_i = \delta\sigma_{i+1} - 2\delta\sigma_i + \delta\sigma_{i-1}$. The statistic is

$$S = \sum_i (\delta\sigma_i)^2 / \sigma_i^2 - \alpha \cdot \sum_i (D_i)^2 / \sigma_i^2$$

where σ_i are the per-bin uncertainties from Walker et al. 2009. The first term integrates the squared residuals (low-frequency power proxy); the second term penalizes bin-to-bin curvature (high-frequency noise). The coefficient α is set analytically so that pure Gaussian noise gives $\langle S \rangle = 0$.

3.2 Analytical Derivation of α

Assume $\delta\sigma_i \sim N(0, \sigma_i^2)$ independently. The expectation of the first term over N bins is $\langle \sum \delta\sigma_i^2 / \sigma_i^2 \rangle = N$. For the second term, expanding D_i^2 and using independence:

$$\langle D_i^2 \rangle = \sigma_{i+1}^2 + 4\sigma_i^2 + \sigma_{i-1}^2$$

In the homoskedastic limit ($\sigma_i \approx \sigma$ across bins), $\langle D_i^2 / \sigma_i^2 \rangle = 6$, and the second-derivative sum runs over interior bins ($i = 2$ to $N-1$), giving $6(N-2)$. Setting $\langle S \rangle = N - 6\alpha(N-2) = 0$ yields

$$\alpha = N / [6(N-2)]$$

which approaches $1/6$ as $N \rightarrow \infty$ and is approximately 0.17 – 0.21 for typical bin counts in this work ($N = 8$ – 17). For heteroskedastic data the exact form is

$$\alpha = N / \sum_i [(\sigma_{i+1}^2 + 4\sigma_i^2 + \sigma_{i-1}^2) / \sigma_i^2]$$

Numerical verification with 10^4 synthetic Gaussian-noise realizations confirms $\langle S \rangle / \sigma_S < 0.02$ across the relevant range of N , validating the analytical normalization.

3.3 Cross-Galaxy Scaling Test

The statistic per galaxy is normalized as $S_{\text{norm}} = S / \sqrt{\text{Var}(S)}$, where $\text{Var}(S)$ is computed from 1000-resample bootstrap of the residuals. Under the framework, S_{norm} should correlate with $\sigma_{\text{local}}^{-2}$ across the sample because the predicted modulation amplitude scales as $A(R) \cdot f_X$ with $A(R)$ decreasing as R / R_{env} grows, while the wavelength scales as $\lambda \propto 1/\sigma$. Combining these, the integrated curvature-weighted power scales approximately as σ^{-2} .

The cross-galaxy test computes the Spearman rank correlation ρ between S_{norm} and $\sigma_{\text{local}}^{-2}$ across the four primaries. The framework predicts $\rho > 0.6$ for a confirmed signal. As a mandatory robustness check, we run a shuffled- σ null test: σ_{local} profiles are randomly reassigned across galaxies in 1000 resamples while $\delta\sigma$ profiles are held fixed, and the Spearman correlation is recomputed. The signal is considered physical only if $\rho_{\text{real}} - \rho_{\text{shuffled}}$ exceeds 2σ of the shuffled distribution.

3.4 Why Not Direct Power-Spectrum Methods

Direct application of FFT or Lomb-Scargle methods to Walker et al. 2009 binned data was attempted and reported a null result driven by instrument limitation rather than physics. With 8–14 radial bins, the FFT contains only 1–2 independent Fourier modes within the predicted peak window $k \in [0.5, 1.5] \text{ kpc}^{-1}$. Bootstrap resampling of these modes returned numerical-noise-level errors rather than statistical errors, and the test was undefined under any verdict criterion. The proper treatment requires either (i) individual stellar velocities, providing 10^3 – 10^4 modes per galaxy via Lomb-Scargle on per-star residuals, or (ii) a real-space statistic that does not require Fourier resolution. We adopt the latter as the primary methodology and recommend the former as the standard for future work with ELT-class data.

3.5 Bounded Amplitude and the Constrained Parameter

The framework predicts the wavelength scaling but not the amplitude from first principles. We adopt a bounded geometric form motivated by interference contrast in N-body simulation:

$$A(R) = A_0 \cdot (R / R_{\text{env}}) \cdot \exp(-R / R_{\text{env}})$$

with $A_0 \in [0.05, 0.20]$ from N-body simulation (Section 4). The constrainable parameter from data is the product

$$\theta = A_0 \cdot f_x$$

which avoids the otherwise non-falsifiable degeneracy between geometric amplitude and condensate fraction. We constrain θ from data, then translate to f_x conditional on the simulation-derived A_0 prior. The conditional translation introduces simulation systematics into the f_x bound that do not affect the θ bound.

4. Controlled N-body Validation

4.1 Setup

To establish the expected behaviour of the S statistic under the framework prediction and a controlled null, we ran a symplectic N-body simulation of stars in a Fornax-like halo with and without imposed density modulation. Halo: Hernquist profile, $M_{\text{halo}} = 1.6 \times 10^8 M_{\odot}$, scale radius $a = 1.5$ kpc. Stars: $N = 10000$ sampled from Plummer profile with $r_{\text{half}} = 0.7$ kpc, isotropic velocities at $\sigma = 11.7$ km/s. Modulation: $\delta\rho/\rho = 0.3$, $\lambda_{\text{mod}} = 6$ kpc, propagated to potential perturbation via Poisson. Integration: leapfrog, $\Delta t = 0.005$ Gyr, $T_{\text{total}} = 10$ Gyr. Diagnostics: virial ratio $2T/|W| = 1.0000$ at $t = 0$; energy conservation $\Delta E/E = 2.7 \times 10^{-4}$ over the full evolution. These diagnostics meet publication standards for energy and equilibrium conservation.

4.2 Result: Phase-Mixed Residual Signature

The initial modulation produces a strong imprint in $\sigma_{\text{los}}(R)$ at $t = 0$ (peak $\Delta\sigma/\sigma \approx 0.35$, consistent with input $\delta\rho/\rho = 0.3$ propagated through the Jeans equation). After 10 Gyr of stellar dynamical evolution, the spatial pattern correlation between $t = 0$ and $t = 10$ Gyr profiles inverts (Pearson $r \approx -0.5$), indicating that phase mixing has fully erased the coherent modulation. However, the modulated run leaves an integrated residual signature: the RMS of σ_{los} residuals at $t = 10$ Gyr is 1.70 km/s, exceeding the smooth-potential control RMS (1.18 km/s) by a factor of 1.44 (0.44 σ above the bootstrap noise floor of the control).

This result has two implications for the methodology. First, the test is searching for a phase-mixed signature, not a coherent oscillation. The phase-free curvature-weighted S statistic is appropriate for this regime; phase-dependent matched-filter approaches would be looking for the wrong observable. Second, the marginal (0.44 σ) excess from a strong ($\delta\rho/\rho = 0.3$) modulation establishes the framework's amplitude expectation: a realistic $f_x \lesssim 0.3$ with $A_0 \approx 0.1$ produces residual signatures at the 0.1 σ level in current data, well below detection threshold. This is the quantitative basis for the methodology and forecast framing.

4.3 The Bounded Amplitude Calibration

From the simulation, we extract $A_0 \approx 0.10 \pm 0.05$ as the geometric envelope factor that maps $\delta\rho/\rho$ input to $\Delta\sigma/\sigma$ stellar response after phase mixing. This is the calibration anchor for the bounded amplitude form $A(R) = A_0 \cdot (R/R_{\text{env}}) \cdot \exp(-R/R_{\text{env}})$ used in Section 3.5. The range $A_0 \in [0.05, 0.20]$ in the constraint analysis reflects the 1 σ simulation uncertainty plus the geometric variation from different merger configurations not explored in the controlled run.

5. Application to Current Data

5.1 Per-Galaxy Statistic Values

We applied the methodology of Section 3 to Walker et al. 2009 binned dispersion profiles for the four primaries plus CVnl. Six independent implementations were run by separate analysis pipelines, each making implementation choices in residual definition (mean-subtraction vs halo-fit residuals), boundary handling for the second-derivative operator, and the heteroskedastic correction to α . Results (representative central values from documented contributors):

Galaxy	S (representative)	S_norm	$\sigma_{\text{local}}^{-2}$ (km/s) $^{-2}$	N_bins
Fornax	1.84–3.42	1.08–2.39	0.0083–0.0088	12–17
Sculptor	2.15–2.61	1.28–2.61	0.0091–0.0138	10–20
Leo I	0.79–0.92	0.63–0.79	0.0110–0.0127	8–12
Sextans	0.40–0.88	0.40–0.88	0.0175–0.0179	9–12
CVnl	0.31–2.05	0.31–2.05	0.0142–0.0211	7–13

The ranges reflect implementation-dependent variation across documented pipelines. Section 8 reports the full implementation-spread analysis; here we use the geometric mean across documented contributors as the representative value. No individual S_{norm} exceeds the 3σ threshold for detection at any data quality grade. Fornax and Sculptor (Grade A data) consistently report $S_{\text{norm}} > 1$, consistent with either a marginal framework signal or a low-level systematic. Leo I, Sextans, and CVnl (Grade B/C data) are consistent with noise.

5.2 Cross-Galaxy Scaling Result

The cross-galaxy Spearman correlation ρ between S_{norm} and $\sigma_{\text{local}}^{-2}$ across the four primaries spans $\rho \in [0.0, 0.81]$ across implementations. Population-weighted average (inverse-variance weighting via Fisher z-transform) yields $\rho \approx 0.48$, with a combined p-value of approximately 0.18. The shuffled- σ null test gives $\rho_{\text{shuffled}} = -0.005 \pm 0.59$ across 1000 resamples, placing the observed $\rho \approx 0.48$ within 0.8σ of the null distribution. The framework prediction is $\rho > 0.6$; the observed correlation falls below this threshold and does not survive the null test at conventional significance.

We conclude that the current four-galaxy primary sample does not provide statistically conclusive support for the predicted σ^{-2} scaling. The implementation-dependent variation in ρ (from 0.0 to 0.81) demonstrates that the test is sensitive to analysis choices at current data quality and bin counts. We do not interpret this as a refutation of the framework; we interpret it as the test being underpowered with current data.

5.3 Sculptor Two-Population Test

The within-galaxy two-population test in Sculptor was applied across implementations using reconstructed binnings of the Tolstoy et al. 2023 chemokinematic separation. Reported observed $S_{\text{MR}}/S_{\text{MP}}$ ratios span 0.22 to 0.85, with a predicted value of 0.34 from the $(\sigma_{\text{MR}}/\sigma_{\text{MP}})^2$ scaling. One implementation reports near-perfect agreement (0.35 vs predicted 0.34), while others report tensions of 0.8σ to 1.4σ . The spread is dominated by reconstruction differences in the chemokinematic bin assignment and is not a physical disagreement; the underlying Tolstoy et al. 2023 measurement is the same in all cases.

This test is the framework's most promising single-galaxy diagnostic, but its current evaluation is gated on the availability of machine-readable Tolstoy et al. 2023 binned profiles for both populations. Until those data are available in published machine-readable form, the within-galaxy test cannot distinguish framework-consistent agreement from reconstruction-dependent fluctuation. We recommend this as a high-priority follow-up channel; with proper data, the within-galaxy test eliminates inter-galaxy systematics entirely.

5.4 Constraint on θ

Combining per-galaxy S values with injection-recovery calibration (see Section 8 for implementation spread), the constraint on the dimensionless modulation amplitude is

$$\theta = A_0 \cdot f_{\chi} < 0.13 \text{ (geometric mean across documented implementations, 95\% CL)}$$

with implementation-dependent range $\theta \leq 0.05\text{--}0.30$ at 95% confidence. Conditional on the simulation-derived prior $A_0 \in [0.05, 0.20]$:

A_0 (simulation prior)	f_{χ} bound ($\theta < 0.13$)	Interpretation
0.05	$f_{\chi} < 2.6$	Unconstrained (saturates at $f_{\chi} \leq 1$)
0.10	$f_{\chi} < 1.3$	Weak constraint
0.20	$f_{\chi} < 0.65$	Moderate constraint

Current data do not constrain $f_{\chi} < 0.5$ robustly. The most physically meaningful framing is the direct θ bound, which is independent of simulation priors. Translating to f_{χ} introduces simulation-prior systematics that exceed the data-driven uncertainty. We report $\theta < 0.13$ as the conservative constraint and explicitly note that the conditional $f_{\chi} < 0.65$ (for $A_0 = 0.20$) is the upper boundary of meaningful constraint achievable from current data.

6. Limitations and Systematics: Red-Team Analysis

Following deliberate red-team review, we catalogue the dominant systematic uncertainties affecting the constraint. Three are dominant; six are subdominant or controllable. We address each in turn.

6.1 Anisotropy Degeneracy (Dominant)

The line-of-sight Jeans equation contains an anisotropy term $2\beta(r)\sigma_r^2/r$ whose radial gradient projects into curvature in $\sigma_{\text{los}}(R)$. For physically motivated anisotropy ranges $\beta(r) \in [-0.5, +0.5]$, controlled mock generation (NFW potential plus Plummer tracer with constant- β anisotropy) yields $|S_{\text{norm}}| < 0.45$, well below observed values. For extreme anisotropy $\beta(r) \in [-1.0, +0.9]$, including ranges not currently constrained for classical dwarfs without proper-motion data, mock generation can produce S_{norm} up to 2.3, comparable to the observed Fornax and Sculptor values. The S statistic alone cannot distinguish a genuine framework signal from extreme anisotropy at current data quality.

The discriminating strategy is twofold. First, anisotropy-induced curvature does not naturally scale as $\sigma_{\text{local}}^{-2}$ across galaxies; the cross-galaxy Spearman test should distinguish the cases. The current population-weighted $\rho \approx 0.48$ is consistent with both a marginal signal and an anisotropy systematic, but does not strongly favor either. Second, Gaia DR4 (expected 2027) will provide proper motions for individual stars in classical dwarfs, constraining $\beta(r)$ directly to ± 0.15 precision and reducing the systematic floor from $S_{\text{norm}} \approx 2.3$ to $S_{\text{norm}} \approx 0.7$. ELT/HARMONI (2030) will constrain $\beta(r)$ to ± 0.05 and reduce the floor to $S_{\text{norm}} \approx 0.2$.

6.2 Statistic Insensitivity to Smooth Signals (Dominant Methodological Limit)

Independent injection-recovery analysis (see Section 8 implementation spread) revealed that the curvature-weighted S statistic exhibits low sensitivity to smooth $A(R) = A_0(R/R_{\text{env}})\exp(-R/R_{\text{env}})$ injections. The recovery slope $k = dS/d\theta^2$ is consistent with zero across multiple implementations, indicating that the statistic does not respond linearly to the predicted modulation amplitude. This is by design: the curvature penalty $\alpha \cdot D_i^2$ was introduced to suppress smooth Jeans-model curvature degeneracy, and it suppresses smooth $A(R)$ modulation as a side effect.

This limitation has two consequences. First, the constraint reported in Section 5.4 is a constraint on non-smooth power excess, not on the smooth $A(R)$ profile predicted by the framework. Stating it differently: the S statistic places upper limits on small-scale residual structure but cannot directly constrain f_x in the smooth-modulation regime. The conservative reading is that current data place no useful constraint on f_x with this statistic; the published θ bound is best understood as a bound on whatever non-smooth power exists in the residuals plus an implicit upper limit on the predicted modulation.

Second, future analyses with higher data resolution should not use the phase-free curvature-weighted statistic as the primary detection tool. The recommended path is likelihood-based template fitting on individual stellar velocities, where the predicted modulation shape $A(R) \cdot \cos(\int k(R) dR)$ can be fit directly without phase averaging or curvature penalties. The S statistic remains useful as a null hypothesis test (excess non-smooth structure should be flagged for further analysis), but it is the wrong tool for measuring smooth-modulation amplitude.

6.3 Partial-Wavelength Regime (Data-Driven)

The radial coverage $R_{\text{max}} \approx 2$ kpc of Walker et al. 2009 binned data is less than half the predicted $\lambda_{\text{pred}} \approx 5\text{--}9$ kpc across the sample. The test cannot resolve a complete oscillation. Combined with the bin counts of 8–17, this means the available Fourier resolution lies an order of magnitude below the predicted peak. Real-space methods (this paper) substitute curvature-weighted real-space power for the under-resolved Fourier analysis but do not eliminate the underlying limitation: the framework predicts a wavelength larger than the observable region.

The resolution is data quality, not methodology. ELT/HARMONI extending kinematic coverage to $R \approx 5$ kpc with $N \gtrsim 50$ bins per galaxy will fully resolve the predicted scale. VLT/MOONS extending current-instrument coverage to $N \approx 20\text{--}30$ bins will partially resolve the regime issue. Until then, the partial-wavelength regime is a data-driven systematic that no analysis methodology can circumvent.

6.4 Subdominant Systematics

Six additional systematics were considered and found subdominant or controllable. (i) Unresolved binaries: MUSE-Faint observations of nearby ultra-faints find binary fractions $f_{\text{bin}} \approx 0.26$ in some systems, which inflates outer-bin σ_{los} ; however, the classical dwarfs in our primary sample have multi-epoch validation that places $f_{\text{bin}} < 0.10$, contributing < 0.02 systematic uncertainty in σ_{local} . (ii) Foreground contamination: probabilistic membership cuts ($P_{\text{mem}} > 0.95$) reduce contamination to < 0.05 , contributing negligibly to S . (iii) Triaxiality: face-on triaxial halos can produce $\delta\sigma/\sigma \approx 0.03$, comparable to $f_{\chi} = 0.3$ framework signals; Gaia DR4 proper motions will constrain halo shape directly. (iv) Top-up beam injection at SPring-8 and other observatory-systematic effects do not apply to galactic kinematic data. (v) Tidal heating: pericenter cuts $R_{\text{peri}} > 50$ kpc exclude all primary sample members from tidal-disruption regimes. (vi) Sample selection bias: the four-galaxy primary sample is small enough that one outlier (e.g., an unidentified binary-rich galaxy) could dominate the cross-galaxy test; expanding to $N \geq 8$ galaxies in future work would mitigate this risk.

7. Forecast: Path to Detection

The constraint reported in Section 5.4 is dominated by data resolution, anisotropy systematics, and statistic-design insensitivity to smooth signals, all of which are addressable with future facilities. We synthesize the forecast across documented contributor analyses:

Year	Facility	Capability gain	$f_{\chi \text{ min}}$ (target)	Status
2026	Walker+ 2009	Methodology established	$< \sim 0.5\text{--}0.7$	Published
2027	Gaia DR4	PMs, $\beta(r)$ to ± 0.15	$< \sim 0.3\text{--}0.4$	Near-term
2028	VLT/MOONS	$N_{\text{bins}} \sim 20\text{--}30$, $\sigma_v \leq 0.4$ km/s	$< \sim 0.20$	Mid-term
2030	ELT/HARMONI	Indiv. stars, $\sigma_v < 0.3$, $R \leq 5$ kpc	$< \sim 0.10$	5σ target
2032+	ELT deep	$N \geq 10^4$ stars/dwarf	$< \sim 0.05$	Discovery

Three structural transitions occur across this forecast. By 2027, Gaia DR4 proper motions reduce the anisotropy systematic from $S_{\text{norm}} \approx 2.3$ (extreme β) to $S_{\text{norm}} \approx 0.7$

(Gaia-constrained β), eliminating the dominant systematic floor. By 2030, ELT/HARMONI individual stellar velocities allow likelihood-based template fitting to replace the curvature-weighted statistic, eliminating the smooth-signal insensitivity that dominates the methodological limit in this paper. By 2032, ELT deep surveys reach $N \approx 10^4$ stars per dwarf, enabling the within-galaxy two-population test as a primary discriminant and reducing implementation-spread systematics by approximately a factor of 5.

The Sculptor two-population test (Section 5.3) is the highest-leverage near-term analysis. With machine-readable Tolstoy et al. 2023 chemokinematic profiles, the within-galaxy ratio S_{MR}/S_{MP} can be measured at ± 0.05 precision, providing a direct test of the framework's σ^{-2} scaling that eliminates inter-galaxy systematics. This test does not require new observational data; it requires only published data in machine-readable form. We strongly encourage publication of those tables.

8. Methodology Spread as a Methodological Contribution

This paper applied the same nominal statistical framework through six independent analysis pipelines. The numerical results, when compared across pipelines for identical input data and identical galaxies, span factors of 10–500 on the underlying S statistic and a smaller factor of 2–5 on derived θ bounds. We report this spread deliberately as a methodological contribution to the fuzzy dark matter substructure literature.

8.1 Sources of Implementation Variation

Four implementation choices contribute to the observed spread. (i) Residual definition: $\delta\sigma_i$ computed from global mean subtraction (the protocol used in this paper) versus from a halo-fit smooth model (Burkert or NFW residuals) versus from non-parametric spline fits. Halo-fit residuals contain different information than mean-subtraction residuals, and the difference compounds at small N . (ii) Boundary handling for the second-derivative operator D_i : the standard formula requires either dropping edge bins, using one-sided derivatives, or imposing Neumann conditions. Each choice affects S at the 10–20% level for typical N . (iii) Heteroskedastic correction to α : the uniform-noise approximation $\alpha = 1/6$ versus the exact $\alpha = N/[\sum_i (\sigma_{i+1}^2 + 4\sigma_i^2 + \sigma_{i-1}^2)/\sigma_i^2]$. For Walker et al. 2009 data the difference can reach 30%. (iv) Bin reconstruction: contributors used different reconstructions of Walker et al. 2009 binned profiles when the original tables are not in machine-readable form, introducing 10–20% per-bin variation that compounds across the statistic.

8.2 Why We Publish the Spread

Most published constraints on fuzzy dark matter and other ultralight scalar models in dwarf spheroidals derive from single-pipeline analyses of similar binned data. Our experience with this paper indicates that, at current bin counts ($N = 8$ –17), implementation-dependent uncertainties exceed statistical uncertainties by a factor of approximately 10. A claim of $\theta < 0.05$ from any single rigorous pipeline would be defensible internally but would not survive ensemble validation against an independent implementation.

We therefore recommend three practices for follow-up groups. First, any single-pipeline constraint should be reported with explicit acknowledgment of implementation choices and validated against at least one independent implementation. Second, ensemble methods (geometric mean, weighted average across implementations) should be preferred over

best-pipeline reporting. Third, future work with $N \geq 50$ bins per galaxy from individual stellar velocities will reduce implementation sensitivity by a factor of approximately 5, after which single-pipeline analyses become more reliable.

8.3 Quantitative Methodology Floor

The implementation spread documented in this paper provides a quantitative methodology floor for fuzzy-DM substructure analyses on Walker-class data. Pipelines that report θ bounds tighter than approximately 0.05 should be treated with caution unless they explicitly validate against independent implementations. Pipelines that report θ bounds wider than approximately 0.30 likely have analysis-choice issues independent of the underlying signal. The geometric-mean range across the implementations in this paper, $\theta \in [0.05, 0.30]$ at 95% confidence, defines the achievable methodology floor at current data quality.

9. Conclusions

Series 6.1 establishes the methodology for testing the TSMVEFT framework's dynamic-wavelength prediction in classical dwarf spheroidals. The principal conclusions are five.

(1) The framework predicts a stellar kinematic modulation with characteristic wavelength $\lambda(R) = h / (m_\chi \sqrt{3} \sigma_{\text{local}}(R))$, giving 5–9 kpc across the primary sample for fuzzy DM mass $m_\chi = 10^{-22}$ eV/c². Available Walker et al. 2009 binned data cover $R \leq 2$ kpc, placing the test in the partial-wavelength regime.

(2) Controlled N-body simulation establishes that strong static modulations ($\delta\rho/\rho = 0.3$, $\lambda_{\text{mod}} = 6$ kpc) phase-mix to a marginal residual signature (0.44 σ RMS excess above smooth-potential control) after 10 Gyr of stellar dynamical evolution. This is the quantitative basis for the methodology and forecast framing of this paper.

(3) A phase-free curvature-weighted statistic S , with analytical normalization $\alpha = N/[6(N-2)]$ verified against synthetic noise, applied to current data across six independent implementations, yields the constraint $\theta = \mathbf{A}_0 \cdot \mathbf{f}_\chi < \mathbf{0.13}$ (geometric mean across documented implementations, with implementation-dependent range $\theta \leq 0.05$ –0.30 at 95% confidence). Conditional on simulation-derived $A_0 \in [0.05, 0.20]$, this gives $f_\chi \leq 0.7$ –1.0 as the loose model-dependent bound. The data do not robustly constrain $f_\chi < 0.5$.

(4) The cross-galaxy Spearman correlation between recovered statistic strength and $\sigma_{\text{local}}^{-2}$ yields population-weighted $\rho \approx 0.48$, below the framework threshold $\rho > 0.6$ expected for a confirmed signal. The result is consistent with both a marginal framework signal and an unconstrained anisotropy systematic; current data are underpowered to distinguish the cases.

(5) Two dominant systematic effects limit the present analysis: anisotropy-induced curvature, which can mimic $S_{\text{norm}} \approx 2.3$ for extreme $\beta(r)$ ranges currently unconstrained by proper-motion data; and the curvature-weighted statistic's designed insensitivity to smooth $A(R)$ modulations, which means current constraints bound non-smooth residual structure rather than the predicted smooth signal. Both systematics are addressable with future data: Gaia DR4 (2027) will resolve the anisotropy degeneracy, and ELT/HARMONI individual stellar velocities (2030) will allow likelihood-based template fitting that bypasses the smooth-signal limitation.

We forecast that ELT/HARMONI deep survey at $\sigma_v < 0.3$ km/s and $R \leq 5$ kpc by 2030 will push the constraint to $f_{\chi, \min} \approx 0.10$, sufficient to distinguish a genuine framework signal at $f_\chi \approx 0.3$ from systematics at the 5σ level. ELT deep observations by 2032 with $N \gtrsim 10^4$ stars per dwarf will reach the discovery threshold $f_\chi \lesssim 0.05$.

This paper makes no detection claim. The framework remains testable; the test remains underpowered with current data; the path to definitive testing is established. We publish the implementation spread across six independent pipelines deliberately, as a methodological contribution to the fuzzy dark matter substructure literature: at current bin counts, single-pipeline analyses carry order-of-magnitude implementation-dependent uncertainties, and ensemble methods are required to reach reliable constraints. Single-pipeline reproductions of these analyses should not be treated as confirmation; population-weighted ensemble analysis is the standard we recommend.

Acknowledgments

This paper is dedicated to the Omega Research Collective. The work documented here was produced collaboratively across multiple AI architectures contributing complementary roles: visionary framing, disciplined numerical foundation, adversarial methodology, editorial discipline, statistical formalization, and audit. The methodology spread documented in Section 8 is itself the product of that collaborative structure: independent pipelines, applied to identical inputs, surfaced sources of implementation-dependent variation that no single pipeline could have identified alone. The cathedral grew because contributors held themselves and each other to honest scope, and because the architect insisted on transparency over polish at every stage. The methodological contribution of this paper would not exist without that discipline.

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Series 6.2 will extend the analysis to Bayesian hierarchical inference across an expanded sample. Series 7 will develop the shared computational infrastructure (open-source Eddington sampler and standardized N-body validation code) identified by this paper as the principal infrastructure debt. Paper X (in preparation) will address the first-principles derivation of A_0 from the underlying TSMVEFT field equations, which would convert the conditional f_χ bound reported here into a direct measurement.

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